

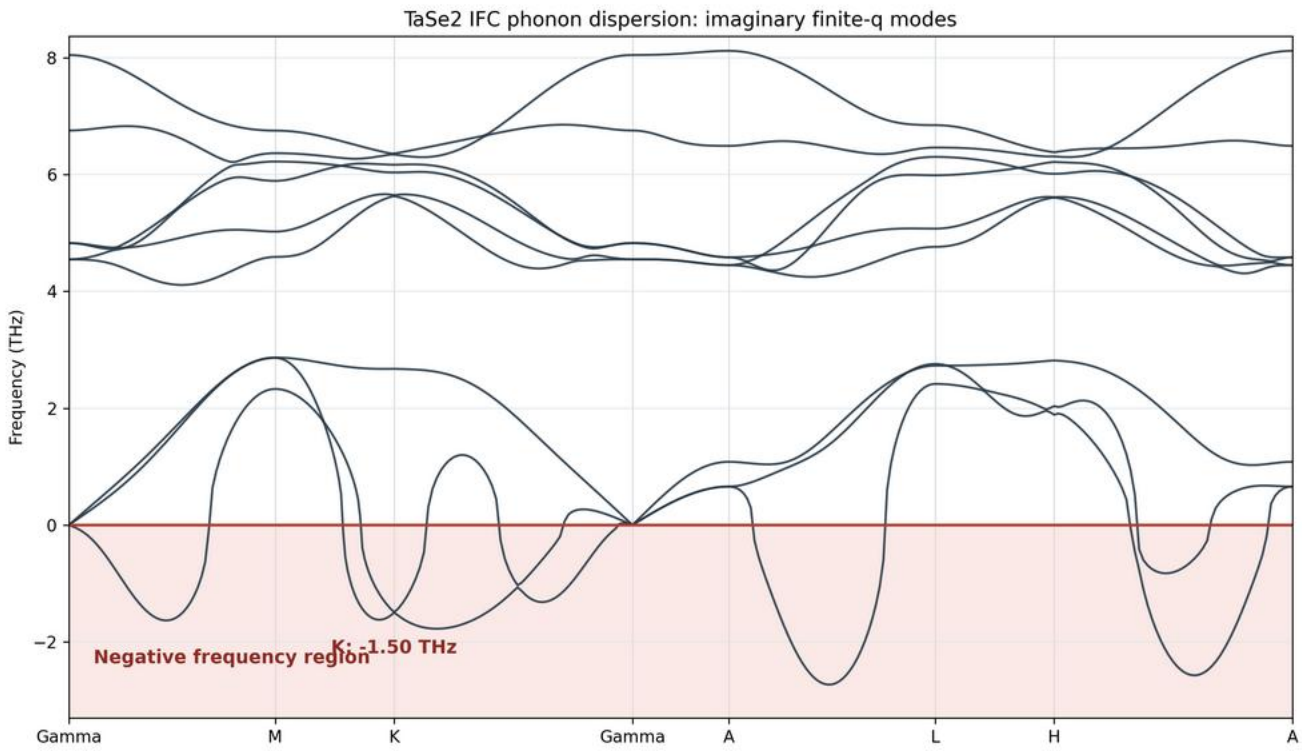
TaSe₂: finite-q imaginary phonons

K: -1.5015 THz

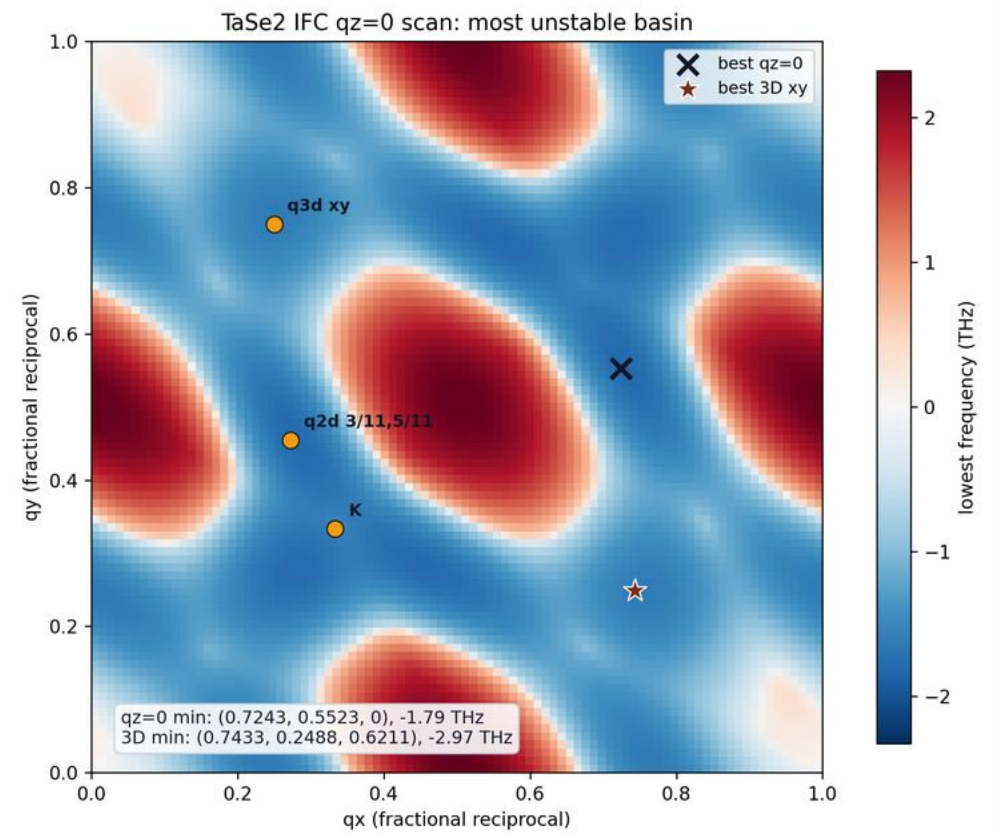
qz=0 min: -1.793 THz

3D min: -2.967 THz

1) dispersion



2) q scan



Unstable q points define candidate CDW supercells

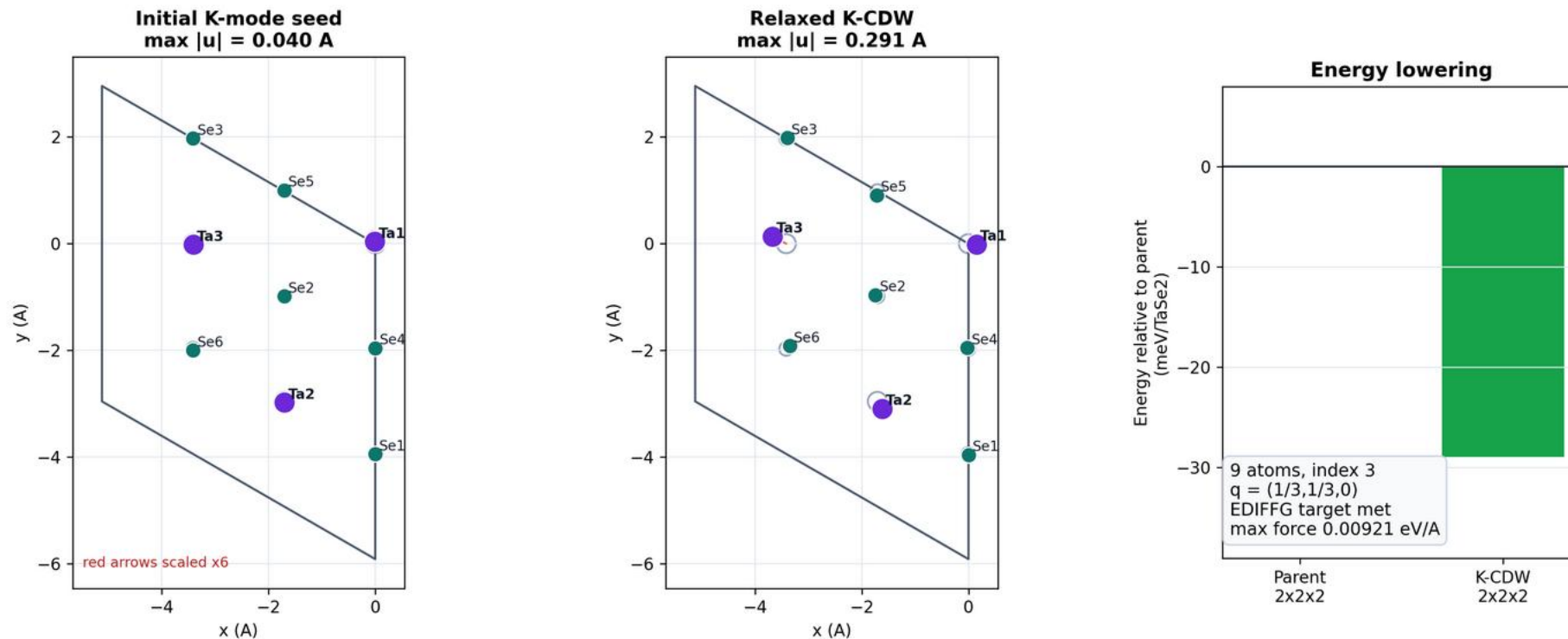
TaSe2 CDW candidate supercells: q folding, seed distortions, and calculation status

candidate	q	IFC min THz	index	atoms	matrix rows	seed displacement	reason / status
K 1/3,1/3,0	(0.3333, 0.3333, 0)	-1.501	3	9	[[-1,1,0]; [-1,-2,0]; [0,0,1]]	mode 0 phase0 + phase90 max u = 0.04 Å	Compact K-CDW reference. DECISIVE: phase0 dense 2x2x2 relaxed
q3d 1/4,3/4,3/8	(0.25, 0.75, 0.375)	-2.964	8	24	[[0,-1,2]; [-4,0,0]; [-1,-2,2]]	mode 0 phase0 + phase90 max u = 0.04 Å	Approximates strongest 3D IFC minimum. Supporting: phase0 Gamma relax only
q2d 3/11,5/11,0	(0.2727, 0.4545, 0)	-1.792	11	33	[[-1,5,0]; [-1,-6,0]; [0,0,1]]	mode 0 phase0 + phase90 max u = 0.04 Å	Closest compact qz=0 minimum approximant. Prepared: phase0/phase90 seeds
q2d 2/7,3/7,0	(0.2857, 0.4286, 0)	-1.786	7	21	[[2,1,0]; [-1,3,0]; [0,0,1]]	mode 0 phase0 + phase90 max u = 0.04 Å	Smaller qz=0 approximant. Prepared: phase0/phase90 seeds
q2d 1/4,1/2,0	(0.25, 0.5, 0)	-1.742	4	12	[[-2,-1,0]; [0,-2,0]; [0,0,1]]	mode 0 phase0 + phase90 max u = 0.04 Å	Very compact strong in-plane instability. Prepared: phase0/phase90 seeds

Seed displacement means the initial eigenvector distortion before VASP relaxation. Only the K phase0 branch is the completed dense-k proof.

Dense K-CDW relaxation lowers the energy

Decisive TaSe2 K-CDW supercell: distortion lowers energy



Relaxation evidence: seed max displacement 0.040 Å $\rightarrow$ relaxed max displacement 0.291 Å; energy decreases by 28.964 meV/TaSe2.

Ta-Ta split: 3.4123 Å uniform $\rightarrow$ 3.0805, 3.0934, 3.3385, 3.3530, 3.3654, 3.3773 Å. Gray open circles are parent positions; colored atoms are displaced positions.

-28.964 meV/TaSe2

relaxed CDW lower than parent

0.04 Å $\rightarrow$ 0.291 Å

seed grows into relaxed PLD

Next check

phonons of relaxed CDW cell